Week 1 – Introduction to Python Programming

Objective

The objective of Week 1 was to understand the fundamental concepts of Python programming and to implement simple Python programs. This week focused on learning Python variables, data types, operators, input/output operations, conditional statements, and loops. These concepts help in building the foundation for writing Python programs for data processing and analysis.

1. Python Variables

Variables are used to store data values in a program. In Python, variables do not require explicit declaration of type. A variable is created automatically when a value is assigned to it.

Example:

temperature = 25

2. Data Types

Python supports different types of data which are used to store various kinds of information.

Some commonly used data types include:

- Integer (int) – used to store whole numbers
- Float (float) – used to store decimal numbers
- String (str) – used to store text
- Boolean (bool) – represents True or False values

3. Operators

Operators are used to perform operations on variables and values. Arithmetic operators are commonly used in Python programs.

Common arithmetic operators include:

- Addition (+)
- Subtraction (-)
- Multiplication (*)
- Division (/)

4. Input and Output

Python provides functions to interact with the user.

- input() function is used to take input from the user.
- print() function is used to display output.

Example:

```
name = input("Enter your name: ")
print("Hello", name)
```

5. Conditional Statements

Conditional statements allow a program to make decisions based on conditions. Python uses if, elif, and else statements for decision making.

Example:

```
if temperature > 30:
    print("It is hot")
else:
    print("Weather is moderate")
```

Program Implemented – 1: Temperature Converter

Description

A Python program was developed to convert temperature between Celsius and Fahrenheit. The program provides two options to the user. The user can choose to convert Celsius to Fahrenheit or Fahrenheit to Celsius.

Conversion Formulas

Celsius to Fahrenheit:
$F = (C \times 9/5) + 32$

Fahrenheit to Celsius:
$C = (F - 32) \times 5/9$

Program Code

```
# Temperature Converter
print("Temperature Converter")
print("1. Celsius to Fahrenheit")
```

```
print("2. Fahrenheit to Celsius")

choice = int(input("Enter choice (1 or 2): "))

if choice == 1:
    celsius = float(input("Enter temperature in Celsius: "))
    fahrenheit = (celsius * 9/5) + 32
    print("Temperature in Fahrenheit:", fahrenheit)

elif choice == 2:
    fahrenheit = float(input("Enter temperature in Fahrenheit: "))
    celsius = (fahrenheit - 32) * 5/9
    print("Temperature in Celsius:", celsius)

else:
    print("Invalid choice")
```

OUTPUT:

```
PS C:\Users\HP\Documents\Data Science Month-1 ( D. Pravallika)\Week - 1> python Temperature_Converter.py
Temperature Converter
1. Celsius to Fahrenheit
2. Fahrenheit to Celsius
Enter choice (1 or 2): 1
Enter temperature in Celsius: 79
Temperature in Fahrenheit: 174.2
```

Program Implemented – 2: Simple Calculator

Description

This program performs basic arithmetic operations such as addition, subtraction, multiplication, and division based on user input.

Program Code

```
# Simple Calculator
a = float(input("Enter first number: "))
b = float(input("Enter second number: "))
op = input("Enter operator (+,-,*,/): ")

if op == '+':
    print("Result:", a + b)
```

```
elif op == '-':
    print("Result:", a - b)
elif op == '*':
    print("Result:", a * b)
elif op == '/':
    print("Result:", a / b)
else:
    print("Invalid operator")
```

OUTPUT:

```
PS C:\Users\HP\Documents\Data Science Month-1 ( D. Pravallika)\Week - 1> python Simple_Calucalator.py
Enter first number: 5
Enter second number: 8
Enter operator (+,-,*,/): /
Result: 0.625
```

Program Implemented – 3: Average Temperature

Description

This program calculates the average of multiple temperature values entered by the user.

Program Code

```
# Average Temperature
temps = list(map(float, input("Enter temperatures: ").split()))
avg = sum(temps) / len(temps)
print("Average Temperature:", avg)
```

OUTPUT:

```
PS C:\Users\HP\Documents\Data Science Month-1 ( D. Pravallika)\Week - 1> python Average_Temperature.py
Enter temperatures: 70 40 20 19
Average Temperature: 37.25
```

Conclusion

Week 1 provided a strong understanding of Python basics including variables, data types, operators, input/output operations, and conditional statements. The implemented programs helped in applying these concepts to real-world problems.